

Geometric Linear Transformations in $\mathbb{R}^3$

1. Rotation

Rotation about the x-axis

The rotation matrix $R_x(\theta)$ for an angle θ is given by:

$$R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$$

Rotation about the y-axis

The rotation matrix $R_y(\theta)$ for an angle θ is:

$$R_y(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$$

Rotation about the z-axis

The rotation matrix $R_z(\theta)$ for an angle θ is:

$$R_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

2. Scaling

The scaling transformation scales the coordinates by factors s_x , s_y , and s_z in the x , y , and z directions, respectively. The scaling matrix S is given by:

$$S = \begin{pmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{pmatrix}$$

3. Reflection

Reflection across the xy-plane

The reflection matrix M_{xy} is:

$$M_{xy} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Reflection across the xz-plane

The reflection matrix M_{xz} is:

$$M_{xz} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Reflection across the yz-plane

The reflection matrix M_{yz} is:

$$M_{yz} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Summary

These matrices can be used to transform points in $\mathbb{R}^3$. For example, if $\mathbf{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ is a vector in $\mathbb{R}^3$, then applying a transformation T can be done by:

$$\mathbf{v}' = T\mathbf{v}$$

where T could be any of the matrices discussed above (rotation, scaling, or reflection).

1. Rotation

Example: Rotate a Point Around the Z-Axis

Let's rotate the point $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ by 90° (or $\frac{\pi}{2}$ radians) around the z-axis.

Rotation Matrix

$$R_z\left(\frac{\pi}{2}\right) = \begin{pmatrix} \cos \frac{\pi}{2} & -\sin \frac{\pi}{2} & 0 \\ \sin \frac{\pi}{2} & \cos \frac{\pi}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Transformed Point

$$\mathbf{v}' = R_z\left(\frac{\pi}{2}\right) \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \cdot 1 + (-1) \cdot 2 + 0 \cdot 3 \\ 1 \cdot 1 + 0 \cdot 2 + 0 \cdot 3 \\ 0 \cdot 1 + 0 \cdot 2 + 1 \cdot 3 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}$$

2. Scaling

Example: Scale a Point

Consider scaling the point $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ by factors $s_x = 2$, $s_y = 3$, and $s_z = 0.5$.

Scaling Matrix

$$S = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0.5 \end{pmatrix}$$

Transformed Point

$$\mathbf{v}' = S \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \cdot 1 + 0 \cdot 2 + 0 \cdot 3 \\ 0 \cdot 1 + 3 \cdot 2 + 0 \cdot 3 \\ 0 \cdot 1 + 0 \cdot 2 + 0.5 \cdot 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \\ 1.5 \end{pmatrix}$$

3. Reflection

Example: Reflect a Point Across the XY-Plane

Let's reflect the point $\mathbf{v} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ across the xy-plane.

Reflection Matrix

$$M_{xy} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Transformed Point

$$\mathbf{v}' = M_{xy} \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix}$$

Homogeneous Coordinates

Definition

For a point $\mathbf{v} = (x, y, z)$ in $\mathbb{R}^3$, the corresponding homogeneous coordinates $[\mathbf{v}]_h$ are given by:

$$[\mathbf{v}]_h = \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

Alternatively, using any non-zero scalar k :

$$[\mathbf{v}]_h = \begin{pmatrix} kx \\ ky \\ kz \\ k \end{pmatrix}$$

Benefits

- Simplifies Transformations: Many geometric transformations can be expressed as matrix multiplications.
- Handles Points at Infinity: Can represent points at infinity by setting the last coordinate to zero.

Transformations in Homogeneous Coordinates

1. Translation

To translate a point by (t_x, t_y, t_z) :

$$T = \begin{pmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

2. Rotation

For rotation around the z-axis by angle θ :

$$R_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

3. Scaling

To scale by factors s_x , s_y , and s_z :

$$S = \begin{pmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Example: Combining Transformations

If we want to apply a translation followed by a rotation and then a scaling, we can combine these transformations into a single matrix M :

$$M = S \cdot R_z(\theta) \cdot T$$

Examples of Combined Transformations in $\mathbb{R}^3$

Example 1: Translate, Rotate, and Scale

Original Point

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Transformations

1. **Translation by $(t_x, t_y, t_z) = (2, 3, 4)$ **:

$$T = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

2. **Rotation around the Z-axis by $\theta = 90^\circ$ **:

$$R_z\left(\frac{\pi}{2}\right) = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

3. **Scaling by $(s_x, s_y, s_z) = (1, 2, 0.5)$ **:

$$S = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Combined Transformation Matrix

$$M = S \cdot R_z\left(\frac{\pi}{2}\right) \cdot T$$

Transformed Point

Convert the point to homogeneous coordinates:

$$[\mathbf{v}]_h = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

Then apply the transformation:

$$\mathbf{v}' = M \cdot [\mathbf{v}]_h$$

Example 2: Scale, Rotate, and Then Translate

Original Point

$$\mathbf{v} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$$

Transformations

1. **Scaling by $(s_x, s_y, s_z) = (2, 2, 2)$ **:

$$S = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

2. **Rotation around the Y-axis by $\theta = 45^\circ$ **:

$$R_y\left(\frac{\pi}{4}\right) = \begin{pmatrix} \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & 0 \\ 0 & 1 & 0 & 0 \\ -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

3. **Translation by $(t_x, t_y, t_z) = (1, 1, 1)$ **:

$$T = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Combined Transformation Matrix

$$M = T \cdot R_y\left(\frac{\pi}{4}\right) \cdot S$$

Transformed Point

Convert the point to homogeneous coordinates:

$$[\mathbf{v}]_h = \begin{pmatrix} 2 \\ 2 \\ 2 \\ 1 \end{pmatrix}$$

Then apply the transformation:

$$\mathbf{v}' = M \cdot [\mathbf{v}]_h$$